

Evidence Report

Two VM Network Commissioning and Connectivity Verification

Item	Detail
Student name	RISA THOMAS KOILA
Student ID	C11/26/FCDF/17156
Course	WADF105 Network Security Fundamentals, IC DFA
Date completed	6 October 2026
Virtual machines	Ubuntu client (icdfa-nslab-client-v1) and OPNsense 26.7 firewall (icdfa-nslab-firewall-v1)
Platform	Oracle VirtualBox; firewall WAN on NAT, LAN and client on Internal Network IC DFA-LAN

1. Laboratory summary

A two-VM network was commissioned in VirtualBox. The OPNsense firewall connects to the internet through a NAT adapter (WAN) and serves a protected internal network, IC DFA-LAN (LAN), at 10.10.10.1/24. The Ubuntu client sits only on IC DFA-LAN, obtains its address by DHCP, and reaches the internet through the firewall. The table summarizes the results; the supporting screenshots are in Section 2.

Test	Command	Result observed
1	ping -c 4 10.10.10.1	4 of 4 replies, 0% packet loss, average 0.737 ms.
2	ping -c 4 1.1.1.1	4 of 4 replies, 0% packet loss, average 47.9 ms.
3	getent hosts opnsense.org	Address returned: 2001:1af8:2050:a001:1::1.
4	curl -I https://opnsense.org	HTTP response headers received (HTTP/1.1 301 redirect to https://opnsense.org/).

Addressing	Value
Firewall WAN (em0)	10.0.2.15/24 via DHCP from VirtualBox NAT; upstream gateway 10.0.2.2
Firewall LAN (em1)	10.10.10.1/24 (static); DHCP range 10.10.10.100 to 10.10.10.200
Ubuntu client (enp0s3)	10.10.10.237/24 via DHCP; default gateway 10.10.10.1; DNS 10.10.10.1
Firewall LAN MAC	08:00:27:48:41:99
Ubuntu client MAC	08:00:27:ff:8a:c6

2. Evidence

E1. VirtualBox network settings for OPNsense

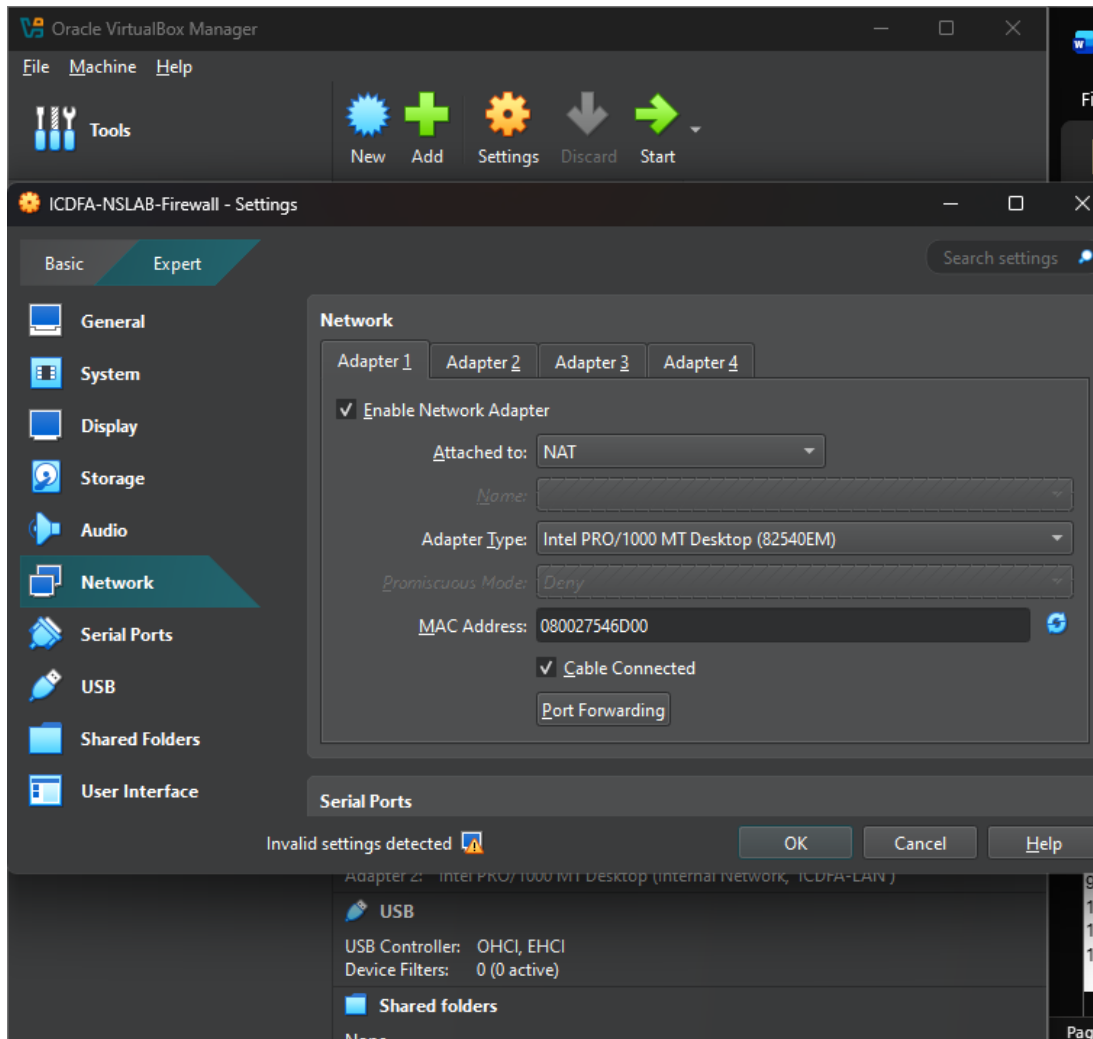


Figure E1a. Adapter 1 of the OPNsense firewall VM is attached to NAT with the cable connected, providing the WAN path to the internet.

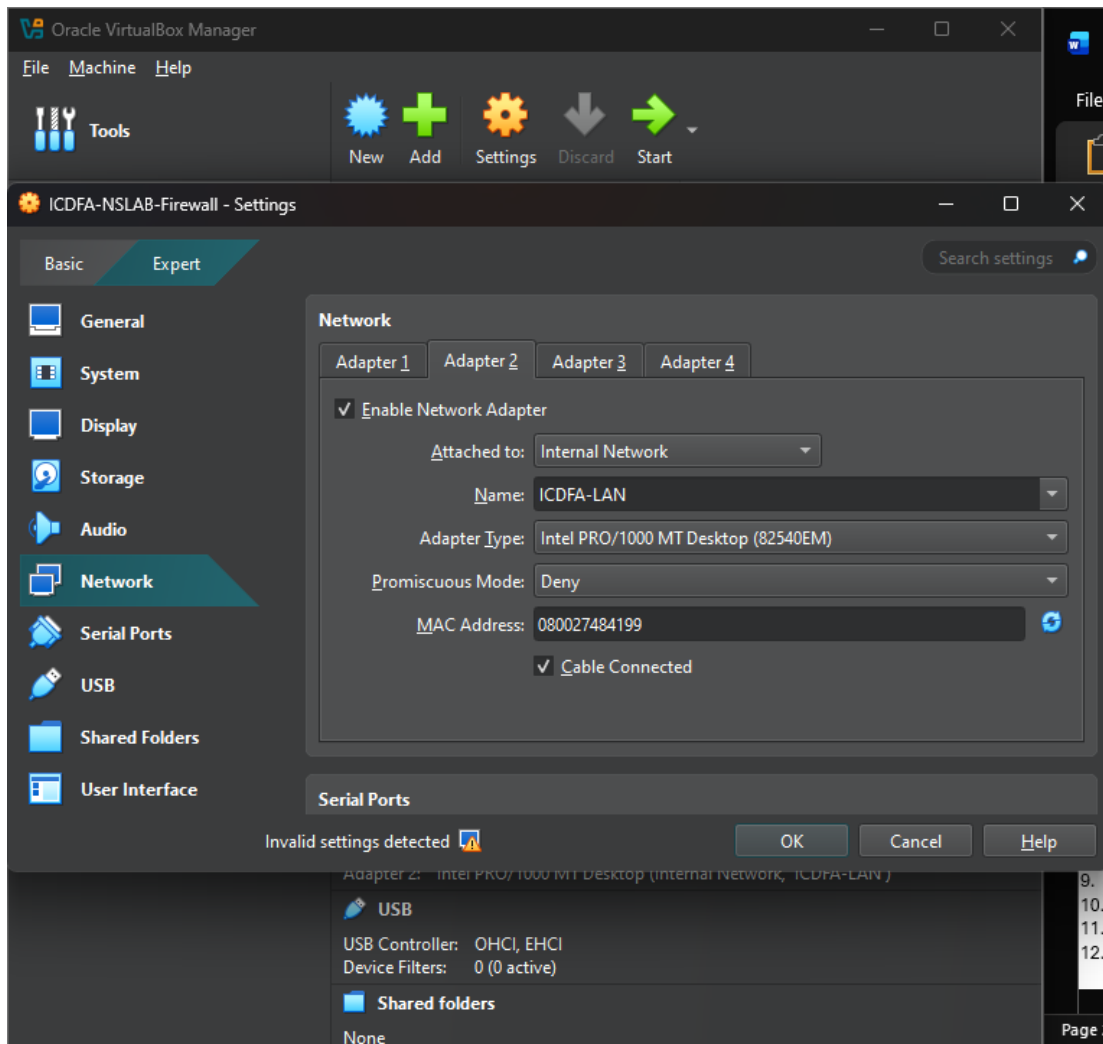


Figure E1b. Adapter 2 of the OPNsense firewall VM is attached to Internal Network ICDFA-LAN with the cable connected, forming the protected LAN.

E2. VirtualBox network settings for Ubuntu

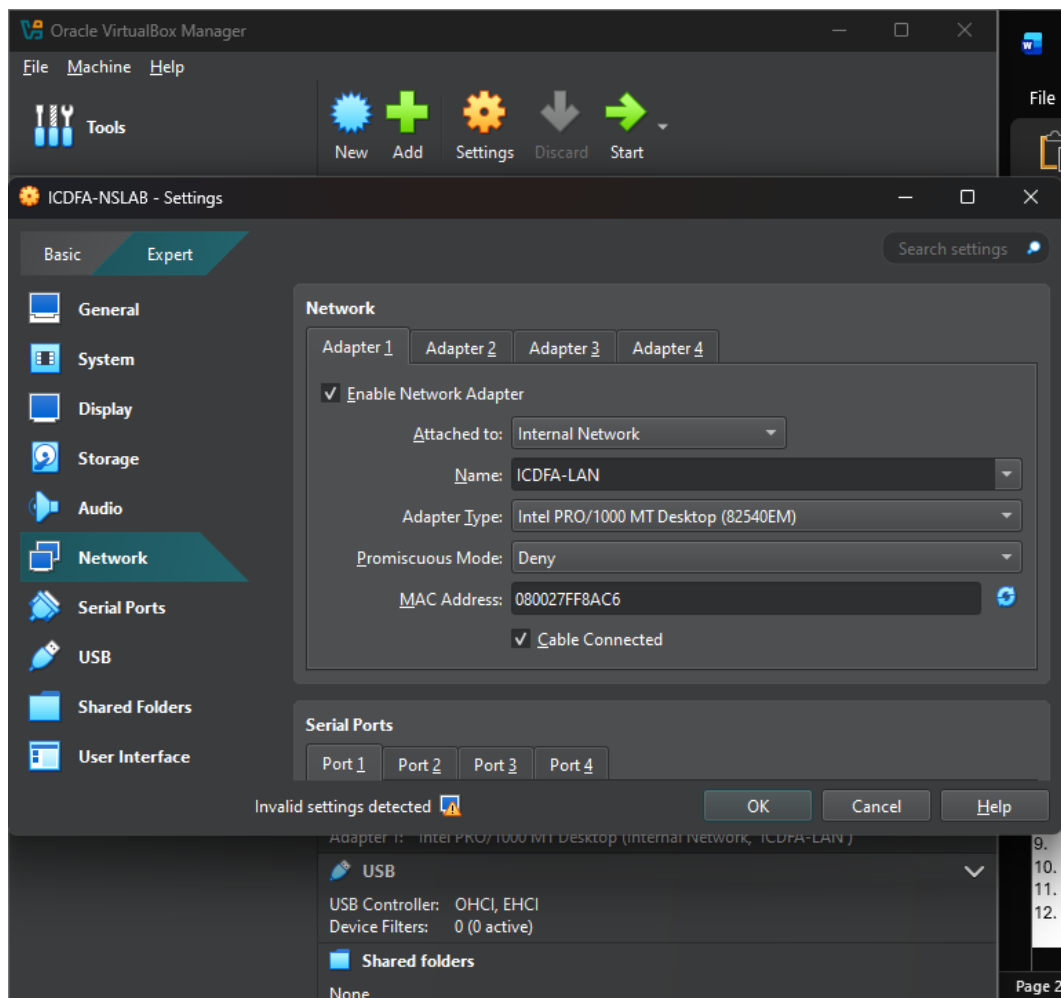


Figure E2. The Ubuntu client has Adapter 1 attached to Internal Network ICDFA-LAN, the same network name as the firewall LAN adapter.

E3. OPNsense WAN and LAN addresses

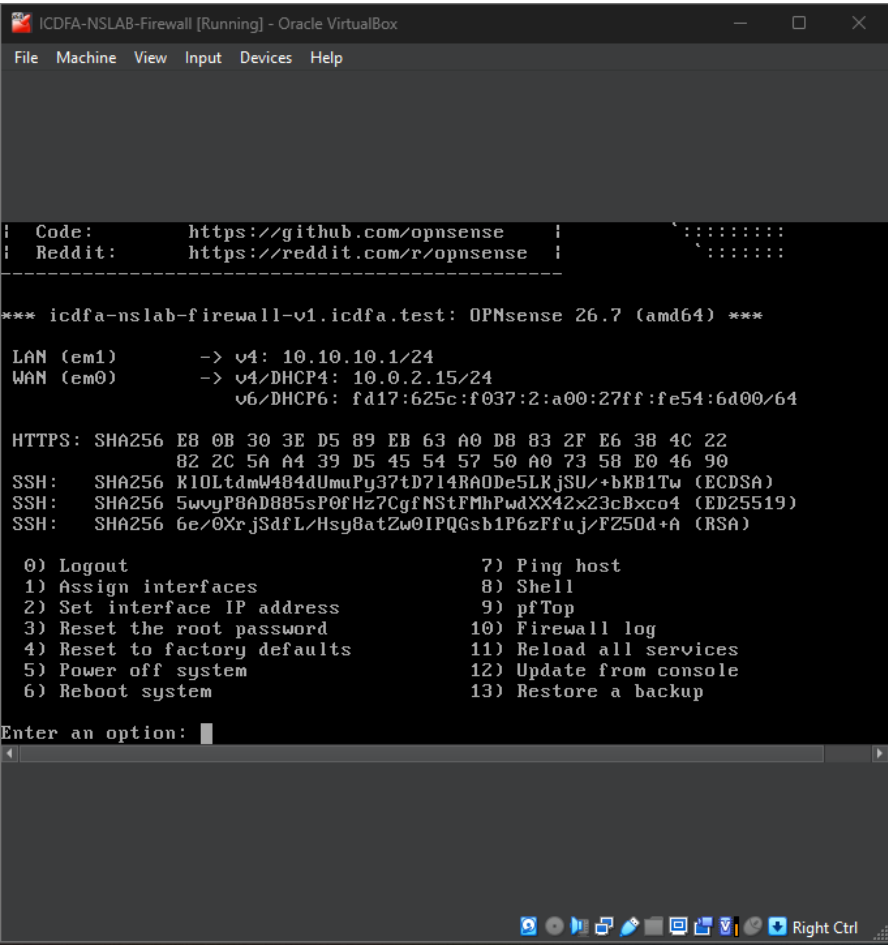


Figure E3a. The OPNsense console shows LAN (em1) at 10.10.10.1/24 and WAN (em0) at 10.0.2.15/24 obtained by DHCP.

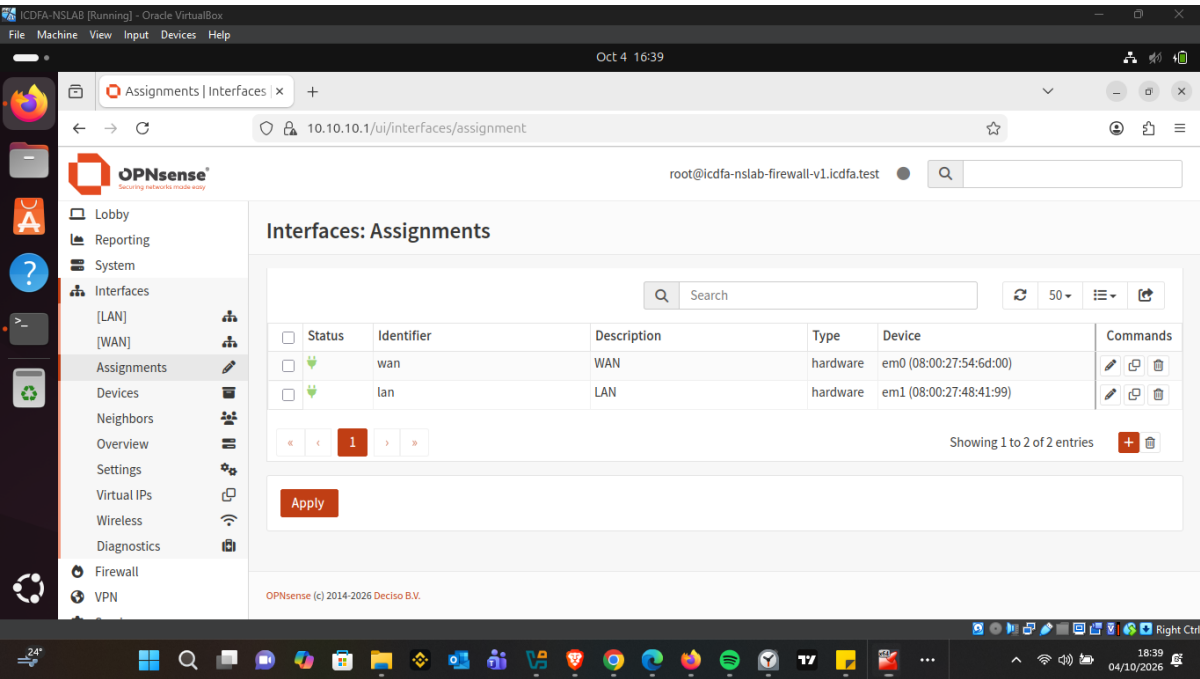
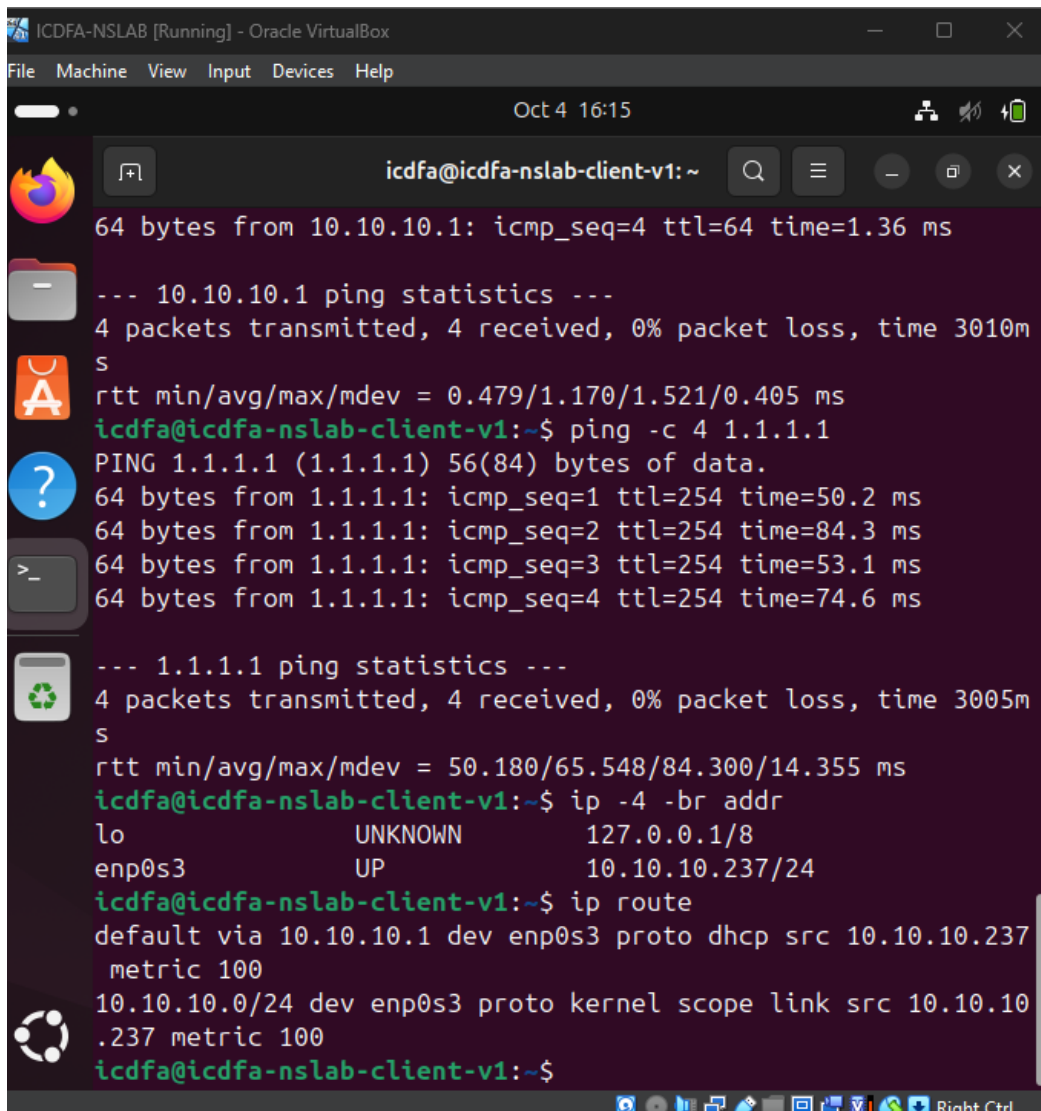


Figure E3b. The OPNsense Interfaces: Assignments page confirms wan is on em0 and lan is on em1, with their MAC addresses.

E4. Ubuntu addressing and routing

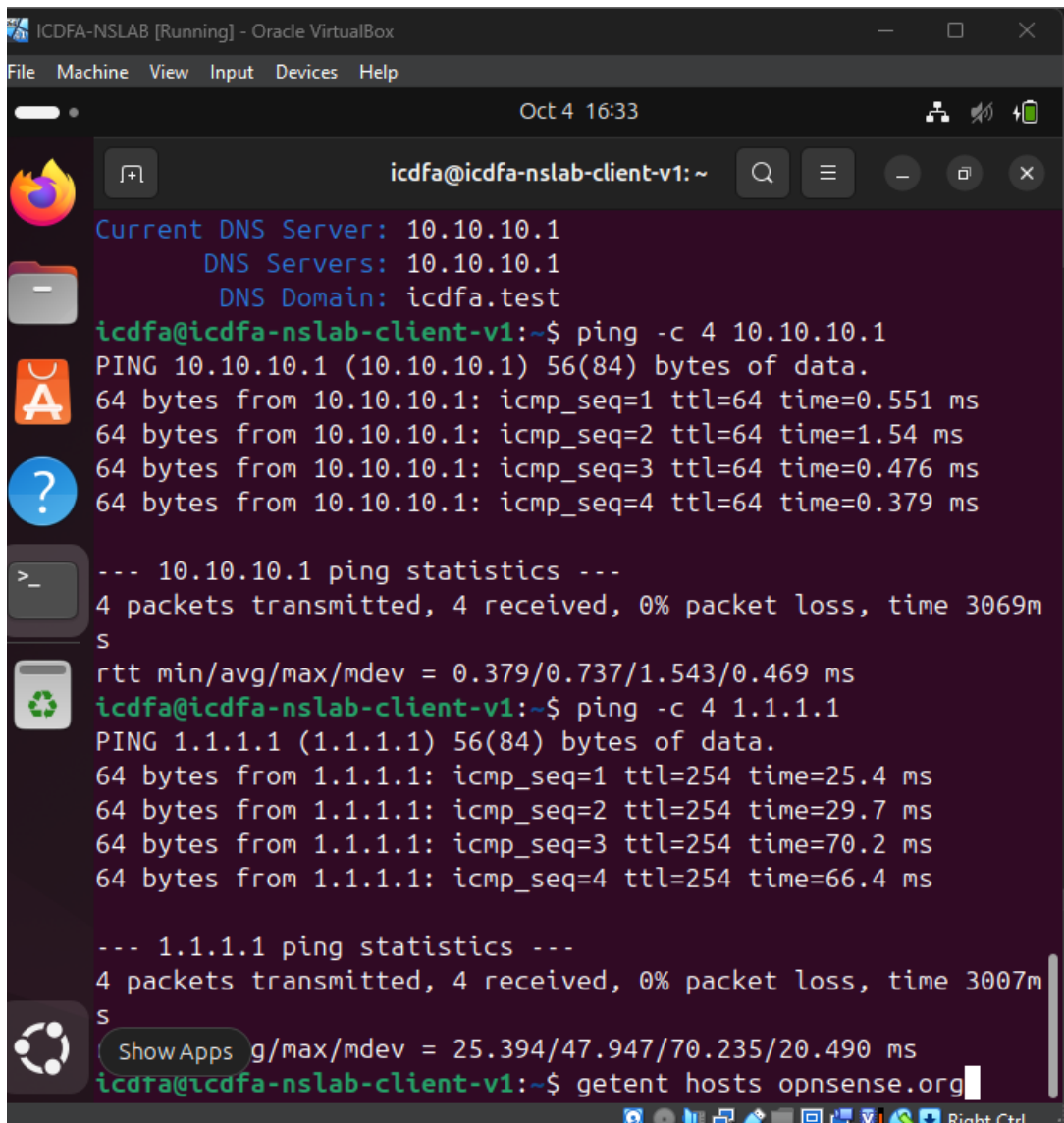


The screenshot shows a terminal window titled 'icdfa@icdfa-nslab-client-v1:~' with the following output:

```
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=1.36 ms
--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3010ms
rtt min/avg/max/mdev = 0.479/1.170/1.521/0.405 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=50.2 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=84.3 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=53.1 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=74.6 ms
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 50.180/65.548/84.300/14.355 ms
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br addr
lo                UNKNOWN          127.0.0.1/8
enp0s3            UP                10.10.10.237/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto dhcp src 10.10.10.237
metric 100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.237 metric 100
icdfa@icdfa-nslab-client-v1:~$
```

Figure E4. The output of `ip -4 -br address` shows `enp0s3` at `10.10.10.237/24`, and `ip route` shows the default route via `10.10.10.1` learned by DHCP.

E5. Gateway, internet and DNS tests



The screenshot shows a terminal window titled 'icdfa@icdfa-nslab-client-v1: ~' within an Oracle VM VirtualBox environment. The terminal displays the following output:

```
Current DNS Server: 10.10.10.1
DNS Servers: 10.10.10.1
DNS Domain: icdfa.test
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.551 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.54 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.476 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.379 ms

--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3069ms
rtt min/avg/max/mdev = 0.379/0.737/1.543/0.469 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=25.4 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=29.7 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=70.2 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=66.4 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 25.394/47.947/70.235/20.490 ms
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
```

Figure E5a. Pings to the gateway 10.10.10.1 and to 1.1.1.1 both succeed with 0% packet loss, proving local reachability, routing and outbound NAT.

```

Oct 4 16:34
icdfa@icdfa-nslab-client-v1: ~
S
rtt min/avg/max/mdev = 0.379/0.737/1.543/0.469 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=25.4 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=29.7 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=70.2 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=254 time=66.4 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007m
S
rtt min/avg/max/mdev = 25.394/47.947/70.235/20.490 ms
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-nslab-client-v1:~$ curl -I https://opnsense.org.
HTTP/1.1 301 Moved Permanently
Date: Sun, 04 Oct 2026 15:33:55 GMT
Server: Apache
X-Redirect-By: WordPress
Strict-Transport-Security: max-age=15552000; includeSubDomain
s; preload
Location: https://opnsense.org/
Content-Type: text/html; charset=UTF-8
Show Apps
icdfa@icdfa-nslab-client-v1:~$

```

Figure E5b. *getent hosts opnsense.org* returns an address and *curl -I https://opnsense.org* returns HTTP headers, proving DNS resolution and web connectivity.

E6. Wireshark ARP, ICMP and DNS

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.10.10.237	10.10.10.1	ICMP	98	Echo (ping) request id=0x21f8, seq=1/256, ttl=64 (reply 1)
2	0.000490845	10.10.10.1	10.10.10.237	ICMP	98	Echo (ping) reply id=0x21f8, seq=1/256, ttl=64 (request 1)
3	1.001810845	10.10.10.237	10.10.10.1	ICMP	98	Echo (ping) request id=0x21f8, seq=2/512, ttl=64 (reply 1)
4	1.002935289	10.10.10.1	10.10.10.237	ICMP	98	Echo (ping) reply id=0x21f8, seq=2/512, ttl=64 (request 1)
5	2.016334757	10.10.10.237	10.10.10.1	ICMP	98	Echo (ping) request id=0x21f8, seq=3/768, ttl=64 (reply 1)
6	2.019642161	10.10.10.1	10.10.10.237	ICMP	98	Echo (ping) reply id=0x21f8, seq=3/768, ttl=64 (request 1)
7	3.016974224	10.10.10.237	10.10.10.1	ICMP	98	Echo (ping) request id=0x21f8, seq=4/1024, ttl=64 (reply 1)
8	3.018857784	10.10.10.1	10.10.10.237	ICMP	98	Echo (ping) reply id=0x21f8, seq=4/1024, ttl=64 (request 1)
9	5.518848624	PCSSystemtec_ff:8a:...	PCSSystemtec_48:41:...	ARP	42	Who has 10.10.10.1? Tell 10.10.10.237
10	5.519945411	PCSSystemtec_48:41:...	PCSSystemtec_ff:8a:...	ARP	60	10.10.10.1 is at 08:00:27:48:41:99
11	14.328740104	10.10.10.237	140.248.169.91	TLSv1.2	93	Application Data
12	14.332879177	140.248.169.91	10.10.10.237	TCP	60	443 → 37840 [ACK] Seq=1 Ack=40 Win=65535 Len=0
13	14.583375819	140.248.169.91	10.10.10.237	TLSv1.2	93	Application Data
14	14.583586180	10.10.10.237	140.248.169.91	TCP	54	37840 → 443 [ACK] Seq=40 Ack=40 Win=65535 Len=0
15	25.793295514	10.10.10.237	34.107.243.93	TLSv1.2	84	Application Data
16	25.795330130	34.107.243.93	10.10.10.237	TCP	60	443 → 34214 [ACK] Seq=1 Ack=31 Win=65535 Len=0

Frame 1: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on
 Ethernet II, Src: PCSSystemtec_ff:8a:c6 (08:00:27:ff:8a:c6), Dst: PCS
 Internet Protocol Version 4, Src: 10.10.10.237, Dst: 10.10.10.1
 Internet Control Message Protocol

Figure E6a. The capture on *enp0s3* shows the ICMP echo request and reply pairs in packets 1 to 8 (red outline) and the ARP request "Who has 10.10.10.1?" with the reply "10.10.10.1 is at 08:00:27:48:41:99" in packets 9 and 10 (blue outline).

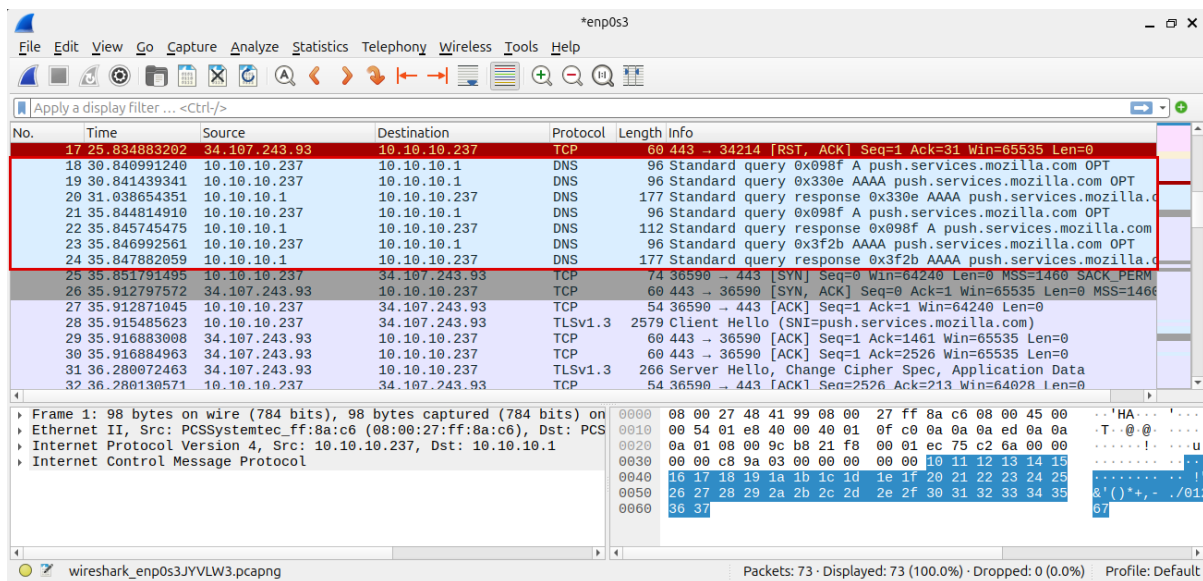


Figure E6b. DNS queries from 10.10.10.237 to the firewall resolver 10.10.10.1 and their matching responses appear in packets 18 to 24 (red outline).

E7. Why the client uses 10.10.10.1 as its default gateway

The Ubuntu client uses 10.10.10.1 as its default gateway because that address is the OPNsense LAN interface, the only router on the 10.10.10.0/24 network. The client can reach other hosts in its own subnet directly, by learning their MAC addresses with ARP. A destination such as 1.1.1.1 is outside that subnet, so the client cannot deliver the packet itself. It sends the packet to the MAC address of the gateway, 10.10.10.1, and the firewall forwards it out of its WAN interface (10.0.2.15), translating the source address with NAT before passing it to the VirtualBox gateway at 10.0.2.2. The firewall hands this gateway address to the client through DHCP, which is why the route table shows "default via 10.10.10.1 dev enp0s3 proto dhcp".

3. Completion questions

1. What is the difference between the OPNsense WAN and LAN interfaces?

The WAN interface (em0) faces the outside, untrusted network. In this lab it is attached to VirtualBox NAT, takes its address by DHCP (10.0.2.15/24) and holds the upstream gateway (10.0.2.2). The LAN interface (em1) faces the protected internal network. It is attached to ICDFA-LAN, has the static address 10.10.10.1/24, serves DHCP to clients, and acts as their default gateway. The firewall routes and applies NAT between the two, and applies its security rules to traffic that crosses them.

2. Why must both internal adapters use the same VirtualBox network name?

A VirtualBox internal network name identifies one virtual Ethernet switch. Adapters with the same name are connected to the same Layer 2 segment, so the client and the firewall LAN can exchange ARP, DHCP and IP traffic. Adapters with different names (the name is case-sensitive) are on separate, isolated switches and cannot communicate, so the client would never reach the firewall or receive a DHCP lease.

3. What information does the default route provide to Ubuntu?

The default route tells Ubuntu where to send any packet whose destination does not match a more specific route. In Figure E4 it names the next hop (10.10.10.1), the outgoing interface (enp0s3), the source address to use (10.10.10.237) and the metric (100), and shows that it was learned by DHCP. Traffic for 10.10.10.0/24 is delivered directly using the connected-network route instead.

4. Which packet exchange allows Ubuntu to learn the firewall MAC address?

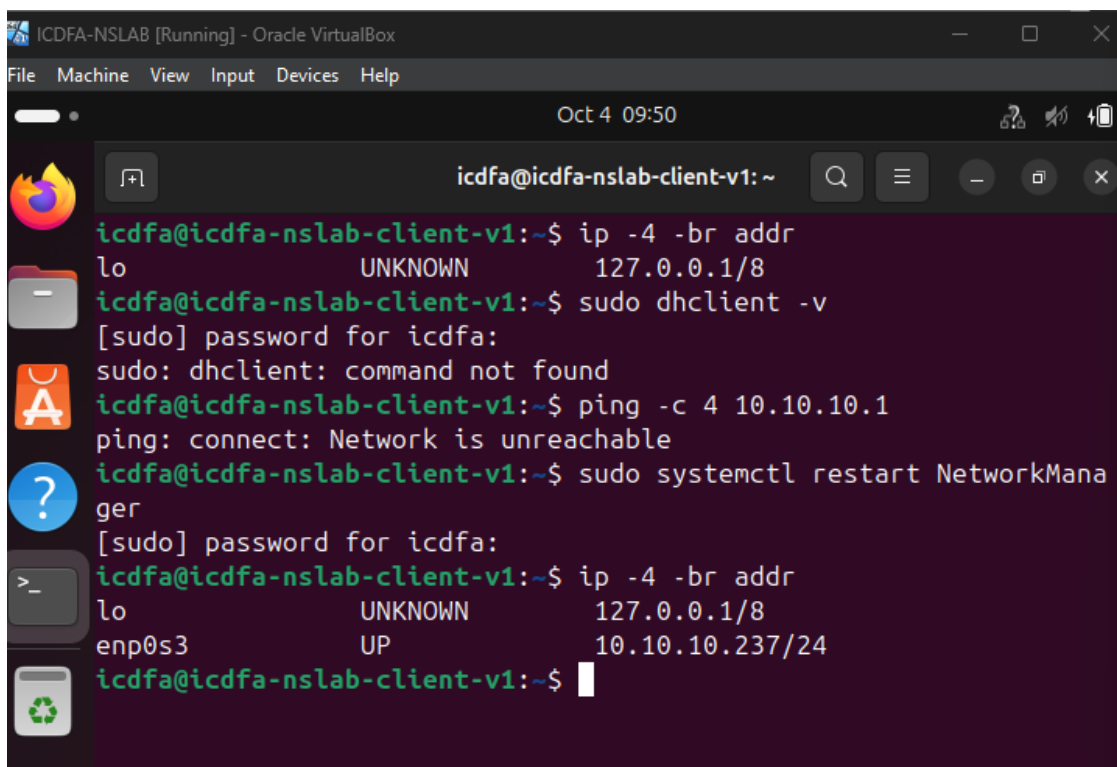
The ARP exchange. Ubuntu broadcasts an ARP request asking "Who has 10.10.10.1? Tell 10.10.10.237", and the firewall answers with a unicast ARP reply stating "10.10.10.1 is at 08:00:27:48:41:99". Both packets are visible in Figure E6a (packets 9 and 10).

5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

Pinging 1.1.1.1 uses a raw IP address, so no name has to be resolved. It proves routing, NAT and internet reachability only. DNS could be misconfigured or unreachable and the ping would still succeed. Resolving a name, as in `getent hosts opnsense.org`, is what tests the DNS service, which is why the lab runs the tests as separate steps.

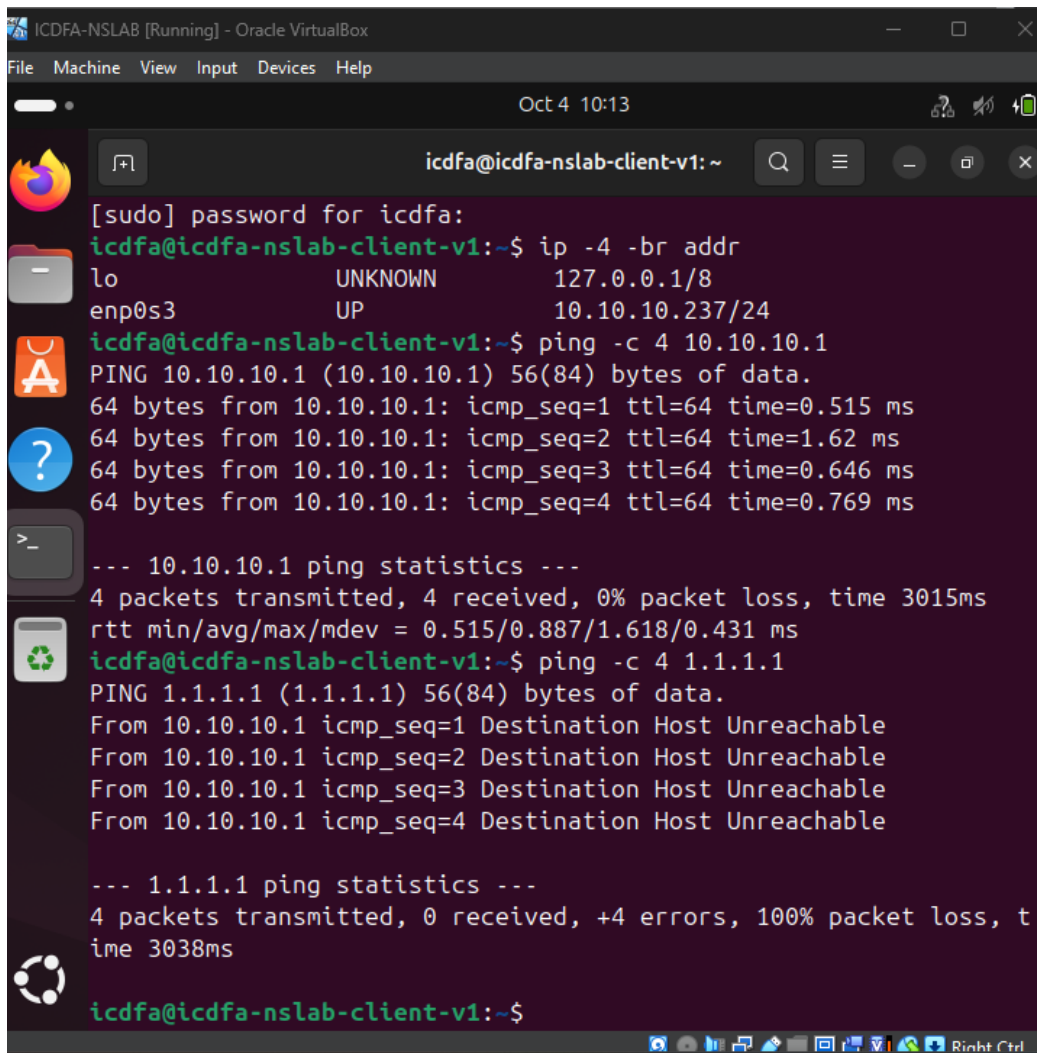
Appendix A. Troubleshooting record

The following screenshots record the faults met during commissioning; in the order they occurred. They are included to show the diagnostic process.



```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br addr
lo                UNKNOWN      127.0.0.1/8
icdfa@icdfa-nslab-client-v1:~$ sudo dhclient -v
[sudo] password for icdfa:
sudo: dhclient: command not found
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
ping: connect: Network is unreachable
icdfa@icdfa-nslab-client-v1:~$ sudo systemctl restart NetworkManager
[sudo] password for icdfa:
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br addr
lo                UNKNOWN      127.0.0.1/8
enp0s3            UP              10.10.10.237/24
icdfa@icdfa-nslab-client-v1:~$
```

Figure A1. The client initially had no IPv4 address on `enp0s3`; `dhclient` was not installed, and restarting `NetworkManager` obtained `10.10.10.237/24`.



The screenshot shows a terminal window titled "ICDFA-NSLAB [Running] - Oracle VirtualBox". The terminal is running on a system named "icdfa@icdfa-ns-lab-client-v1:~". The user has entered the password for "icdfa" and then run the command "ip -4 -br addr". The output shows the loopback interface "lo" with address "127.0.0.1/8" and the ethernet interface "enp0s3" with address "10.10.10.237/24". The user then runs "ping -c 4 10.10.10.1", which succeeds with four successful pings. Finally, the user runs "ping -c 4 1.1.1.1", which fails with four "Destination Host Unreachable" messages. The terminal window has a sidebar with icons for file manager, application store, help, and a terminal icon. The top bar shows the date "Oct 4 10:13" and system status icons.

```
[sudo] password for icdfa:
icdfa@icdfa-ns-lab-client-v1:~$ ip -4 -br addr
lo                UNKNOWN          127.0.0.1/8
enp0s3            UP                10.10.10.237/24
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data:
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=0.515 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.62 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=0.646 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=0.769 ms

--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3015ms
rtt min/avg/max/mdev = 0.515/0.887/1.618/0.431 ms
icdfa@icdfa-ns-lab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
From 10.10.10.1 icmp_seq=1 Destination Host Unreachable
From 10.10.10.1 icmp_seq=2 Destination Host Unreachable
From 10.10.10.1 icmp_seq=3 Destination Host Unreachable
From 10.10.10.1 icmp_seq=4 Destination Host Unreachable

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3038ms

icdfa@icdfa-ns-lab-client-v1:~$
```

Figure A2. The gateway ping succeeded, but the ping to 1.1.1.1 returned "Destination Host Unreachable" from 10.10.10.1, showing the firewall had no usable route to the internet.

```
ICDFA-NSLAB-Firewall [Running] - Oracle VirtualBox
File Machine View Input Devices Help

5) Power off system          12) Update from console
6) Reboot system             13) Restore a backup

Enter an option: 8

root@icdfa-nslab-firewall-v1:~ # netstat -rn | head
Routing tables

Internet:
Destination      Gateway          Flags           Netif Expire
10.0.2.0/24       link#1           U               em0
10.0.2.3          10.0.2.2         UGHS            em0
10.0.2.15         link#3           UHS             lo0
10.10.10.0/24     link#2           U               em1
10.10.10.1        link#3           UHS             lo0
127.0.0.1         link#3           UH              lo0

root@icdfa-nslab-firewall-v1:~ # ping -c 2 10.0.2.2
PING 10.0.2.2 (10.0.2.2): 56 data bytes
64 bytes from 10.0.2.2: icmp_seq=0 ttl=255 time=0.609 ms
64 bytes from 10.0.2.2: icmp_seq=1 ttl=255 time=2.253 ms

--- 10.0.2.2 ping statistics ---
2 packets transmitted, 2 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 0.609/1.431/2.253/0.822 ms
root@icdfa-nslab-firewall-v1:~ #
```

Figure A3. On the firewall, `netstat -rn` showed no default route at this point, while the upstream gateway 10.0.2.2 answered pings, so the WAN link itself was working.

```
ICDFA-NSLAB-Firewall [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Routing tables

Internet:
Destination      Gateway      Flags      Netif Expire
default          10.0.2.2    UGS        em0
root@icdfa-nslab-firewall-v1:~ # nc -zv -w 5 1.1.1.1 443
Connection to 1.1.1.1 443 port [tcp/https] succeeded!
root@icdfa-nslab-firewall-v1:~ # drill opnsense.org @10.0.2.3
;; ->>HEADER<<- opcode: QUERY, rcode: NOERROR, id: 38401
;; flags: qr rd ra ; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
;; QUESTION SECTION:
;; opnsense.org.      IN      A

;; ANSWER SECTION:
opnsense.org.  900      IN      A      89.149.225.137

;; AUTHORITY SECTION:

;; ADDITIONAL SECTION:

;; Query time: 288 msec
;; SERVER: 10.0.2.3
;; WHEN: Sun Oct 4 11:56:15 2026
;; MSG SIZE rcvd: 46
root@icdfa-nslab-firewall-v1:~ #
```

Figure A4. Once the default route via 10.0.2.2 on em0 was in place, the firewall made a TCP connection to 1.1.1.1 port 443 and resolved opnsense.org through the VirtualBox resolver.